Hazard Analysis
Software Engineering

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Table 1: Revision History

Date	**Developer(s)**	**Change**
October 17, 2026	Jacob Brodersen	Added Initial Draft of the Hazard Analysis
October 18, 2026	Jacob Brodersen	Revised Initial Draft of the Hazard Analysis
October 19, 2026	Jacob Brodersen	Finalized Hazard Analysis

Contents

1 Introduction **1**
- 1.1 Background . . . 1
- 1.2 Hazard Definition . . . 1

2 Scope and Purpose of Hazard Analysis **1**
- 2.1 Health . . . 1
- 2.2 Data . . . 2
- 2.3 Education . . . 2
- 2.4 Usability . . . 2

3 System Boundaries and Components **2**

4 Critical Assumptions **3**

5 Failure Mode and Effect Analysis **5**

6 Safety and Security Requirements **14**

7 Roadmap **14**
- 7.1 Safety and Security Requirements for Capstone Timeline . . . 15
- 7.2 Deferred Safety and Security Requirements . . . 16

8 Appendix — Generative AI Use Disclosure **17**
- 8.1 AI Tools Used . . . 17
- 8.2 How and Where Used . . . 17
- 8.3 Verification of AI-Generated Content . . . 17

9 Appendix — Reflection **18**

1 Introduction

1.1 Background

A most perfect magic cube is a cube where each voxel's value is a distinct number from 1 to n^3 and all rows, columns, pillars, 2-dimensional diagonals, 3-dimensional diagonals (triagonals), and broken space diagonals in the cube sum to the same number, also called the magic constant. The order-eight cube is the smallest order most perfect magic cube. Using octal within this order-8 magic cube showcases the magic properties of the cube with the above prescribed lines summing to 4004, half lines summing to 2002, and the complementary numbers (+4 4 4 for the x y z coordinates from a given voxel) summing to 1001 (all in octal).

Our project, Magic 8, is a web-based application that serves as a recreational mathematics game as well as an educational tool for octal addition and subtraction. In this proposed game, players start with one voxel (number) in a most perfect magic cube visible while all other voxels remain hidden. The player must then complete tasks related to finding the voxels belonging to lines that sum to the magic constant. Each round of the game includes eight tasks that increase in difficulty as the player progresses.

1.2 Hazard Definition

A hazard is defined as a system state that if combined with a specific environmental condition, can cause harm/damage. Hazards can result from software defects or incorrectly defined system behavior. These hazards can be categorized as health, data, education, and usability hazards.

2 Scope and Purpose of Hazard Analysis

The purpose of this Hazard Analysis is to identify hazards related to the Magic 8 application and subsequently perform a Failure Mode and Effect Analysis (FMEA) to mitigate the found hazards. Below is the description of the losses that could be incurred because of the hazards. Out of scope features include accessibility concerns, advanced cybersecurity (beyond simply encrypting data and securing passwords), and optimizing or extending the software for outdated or insufficient user hardware or software. Although these considerations will be important for future updates (past the capstone timeline), they do not fit within the scope of a first official version of the application.

2.1 Health

This category is described as the loss of health due to use of the system. This includes both physical health (e.g., photosensitive seizure, see **?**) as well as mental health (e.g., bullying through the application and addiction).

2.2 Data

This category is described as the deletion, corruption, or leakage of data. One loss related to this category is the unauthorized disclosure of private user account information (e.g., password). Another loss is the corruption or deletion of data (both user data as well as game data).

2.3 Education

This category is described as the loss of educational value. This involves harming the user's knowledge of octal addition and subtraction by either teaching them incorrectly or adding to their confusion of the mathematical concept.

2.4 Usability

This category is described as the loss of user engagement, satisfaction, or trust. Any of these losses would reduce the usability of the application and subsequently make it a less valuable product.

3 System Boundaries and Components

The following table describes the main components within the system boundary. Out of scope considerations are described in the Scope and Purpose of Hazard Analysis section.

Table 3: System Components

Component	**Description**
User Database	Stores user account specific data including usernames, emails, passwords, and individual high scores.
User Manager	Handles all aspects of user accounts including account creation, account management, and account deletion.
Competitive / Leaderboard Database	Stores data related to online, competitive statistics such as global high scores, fastest solve times, and rankings.
Competitive / Leaderboard Manager	Handles display and management of competitive statistics and leaderboards.

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Component	Description
Game State Manager	Handles the gameplay for the application. This involves taking in user actions and evaluating them against the current task, progressing through tasks in a round, saving progress of a current game, etc.
Visualization/Rendering Manager	Handles the rendering and projection calculations associated with the 3D visualization. This includes both the magic cube itself and highlighting related to the current task.
Hint/Tutorial Manager	Handles the hints, feedback, and tutorial steps related to the game as well as educational help for octal arithmetic.
User Interface	Handles user input and output for the labels, buttons, and text fields of the application. Essentially, this component handles all user input and output that is not related to rendering the 3D space.
Task Generator	Handles the task generation process for a given round including generating both the problem and solution for a generated task and ramping up difficulty as a round progresses.

Outside of the system boundary are the users themselves and their associated hardware (computer) and web browser. In terms of external interaction, all of the input and output associated with the system goes through the User Interface component with the exception of the 3D visualization output, which is handled by the Visualization/Rendering Manager.

4 Critical Assumptions

The following are critical assumptions that are made about the system. These are necessary so that the scope of the analysis can be focused on realistic and manageable hazards.

One assumption is that users operate modern, standards-compliant web browsers. Another assumption is that external hosting providers meet uptime

targets. A third assumption is that the external services the system uses for data storage have general protections against malicious behaviour, reducing but not removing data risks. Finally, it is assumed that external libraries are free from critical vulnerabilities (however manual checks will still be done to insure the content is reasonable).

5 Failure Mode and Effect Analysis

Below is the Failure Mode and Effect Analysis (FMEA) table. Each row represents an independent failure mode of the system and includes the related component, cause(s) of failure, effect(s) of failure, recommended action(s), and related Safety and Security Requirements.

Table 4: Failure Mode and Effect Analysis (FMEA)

Component	**Failure Modes**	**Cause(s) of Failure**	**Effect(s) of Failure**	**Recommended Action(s)**	**SR**	**Ref.**
User Database	User data exposed to unauthorized individuals	- Weak password storage	- User privacy breach - Loss of trust	- Encrypt sensitive data - Hash based passwords - Conduct security audits	- SR1 - SR2 - PR3 - PR4	H1-a
User Database	User data lost or corrupted	- Error in saving - Database crash	- Loss of user account information	- Automated backups - Transaction logging - Recovery procedures	- SR3 - SR4 - SR11 - PR6 - PR7 - PR8	H1-b
User Manager	Inappropriate and/or harmful usernames	- Inadequate review or control of public usernames	- Decrease in user well-being	- Restrictions on username content	- SR13 - FR44	H2-a

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Component	Failure Modes	Cause(s) of Failure	Effect(s) of Failure	Recommended Action(s)	SR	Ref.
Competitive / Leaderboard Database	Incorrect leaderboard rankings	- Database corruption or loss of data	- User frustration - Loss of user engagement	- Automated backups - Transaction logging - Recovery procedures - Ranking validation	- SR3 - SR4 - SR11 - PR6 - PR7 - PR8	H3-a
Competitive / Leaderboard Database	Unauthorized modification of leaderboard and competitive data	- Insufficient access controls	- User frustration leading to loss of user engagement	- Role-based access control	- SR10 - FR50	H3-b

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Component	Failure Modes	Cause(s) of Failure	Effect(s) of Failure	Recommended Action(s)	SR	Ref.
Competitive / Leaderboard Manager	Incorrect competitive statistics or leaderboard displayed	- Unsynchronized connection to Competitive / Leaderboard Database - Software bug	- Users feel unaccomplished and disatisfied that their achievements are not recognized - Loss of user engagement	- Ensure that Competitive / Leaderboard Manager is up to date with Competitive / Leaderboard Database - Formally verify component software correctly displays the competi-tive/leaderboard data	- SR18	H4-a
Game State Manager	Progress not saved	- Game crash - Failed save	- Loss of user progress leading to frustration and disengagement	- Autosave functionality - Save retries	- SR5 - SR6 - FR30	H5-a

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Component	Failure Modes	Cause(s) of Failure	Effect(s) of Failure	Recommended Action(s)	SR	Ref.
Game State Manager	Incorrect game state loaded	- Corrupted save data - Incorrect state flow	- Correct answer marked incorrect or vice versa - User confusion leading to decreased educational value and disengagement	- Save integrity checks - Validate state flow	- SR7 - FR31	H5-b
Game State Manager	User becomes addicted to the gameplay	- Insufficient reminders of gameplay time	- Addictive or excessive use of the application	- Remind user when they have been playing for an extended period of time	- SR21	H5-c

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Component	Failure Modes	Cause(s) of Failure	Effect(s) of Failure	Recommended Action(s)	SR	Ref.
Visualization / Rendering Manager	Cube or highlighting incorrectly displayed	- Projection error - Visualization / Rendering Manager is relying on an outdated or incorrect game state	- User confusion leading to reduced educational value - Lead user to incorrect decisions	- Formal verification of projection algorithm - Gameplay testing to ensure correct cube and highlighting is displayed - Ensure that the Visualization / Rendering Manager is using the current game state	- SR15	H6-a
Visualization / Rendering Manager	Unstable refresh rate or screen tearing	- Software bug - Incorrect application of animation	- User disorientation leading to disengagement - Risk of photosensitive seizure	- Unit testing of component - Ensure user has minimum hardware specification to sustain refresh rate	- SR14	H6-b

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Component	Failure Modes	Cause(s) of Failure	Effect(s) of Failure	Recommended Action(s)	SR	Ref.
Visualization / Rendering Manager	Excessive and unintended flashing or animation	- Graphics bug - Excessive use of animation	- Photosensitive seizure risk - Eye strain	- Unit testing of component to reduce chance of unintended flashes/animations - User testing of animations to ensure the animations are not excessive - Allow user to disable animations that can pose a risk of photosensitive seizures	- SR19	H6-c

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Component	Failure Modes	Cause(s) of Failure	Effect(s) of Failure	Recommended Action(s)	SR	Ref.
Hint / Tutorial Manager	Incorrect hint provided	- Hints are not properly encoded - Tutorial/hint state flow is incorrect - Hint / Tutorial Manager is not synchronized with game state	- User confusion and frustration - User disengagement - Loss of educational value	- Unit testing of component to ensure correct hint is provided - Validation of hint/tutorial state flow - Ensure that the hint / tutorial manager is synchronized with the current game state	- SR20	H7-a
Hint / Tutorial Manager	Incorrect octal arithmetic provided in a hint	- Octal arithmetic is not properly encoded when generating hints	- Loss of educational value	- Unit testing of component's octal arithmetic - Formal verification of octal arithmetic calculations	N/A	H7-b

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Component	Failure Modes	Cause(s) of Failure	Effect(s) of Failure	Recommended Action(s)	SR	Ref.
User Interface	User input interpreted incorrectly	- Input parsing error (e.g., text fields such as username and email) - User action(s) mapped incorrectly to game command(s)	- Incorrect action taken leading to incorrect game state - User frustration / disengagement - Loss of educational value	- Unit testing of encoded actions - Allow users to confirm parsed data (e.g., confirm user account data before submitting)	- SR16	H8-a
User Interface	User input not registered	- Event handling bug	- Inability to progress - User frustration / disengagement	- Formally validate that all user input leads to an event - Inform user of errors in handling an input event	- SR17	H8-b
Task Generator	Duplicate task generated	- Insufficient shuffling of tasks	- Loss of user engagement - Ineffective education	- Check tasks for uniqueness in a given round - Validate randomness of task generation	- SR8 - FR6	H9-a

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Component	**Failure Modes**	**Cause(s) of Failure**	**Effect(s) of Failure**	**Recommended Action(s)**	**SR**	**Ref.**
Task Generator	Invalid task generated	- Logic error - Error in task generation algorithm	- Unsolvable puzzle leading to disengagement - User confusion / loss in educational value	- Validate solvability of task before giving it to the user	- SR9 - FR25	H9-b
Task Generator	Generated task has incorrect expected answer	- Logic error	- Correct answer is marked incorrect and an incorrect answer could be marked correct - Loss of educational value - User frustration and disengagement	- Validate the generated solution before giving the task to the user	- SR12 - FR25	H9-c

6 Safety and Security Requirements

This section outlines the Safety and Security Requirements that were developed from the FMEA table.

SR1: The system shall encrypt sensitive user data.
SR2: The system shall store user account passwords as cryptographic hashes.
SR3: The system shall automatically generate a backup for each database.
SR4: The system shall log all transactions for each database.
SR5: The system shall automatically save game progress every time there is a change in game state.
SR6: The system shall attempt to save the game state three times before failing and notifying the user.
SR7: The system shall ensure that a save state is valid before loading it as the current game state.
SR8: The system shall ensure that each generated task has not already been generated in the current round.
SR9: The system shall ensure that each generated task is solvable.
SR10: The system shall only allow authorized users to modify competitive statistics and leaderboard records.
SR11: The system shall restore database data from the most recent, valid backup in the event of database corruption or data loss.
SR12: The system shall ensure that a generated solution correctly solves its related generated task.
SR13: The system shall ensure that public usernames meet guidelines for user conduct.
SR14: The system shall warn the user if their hardware specification does not meet minimum requirements to run the application.
SR15: The system shall ensure that the rendered cube and highlighting is consistent with the current game state.
SR16: The system shall allow users to confirm parsed data before submitting.
SR17: The system shall inform the user if an input event is not recognized.
SR18: The system shall ensure that the displayed competitive/leaderboard is up to date (synchronized with the Competitive / Leaderboard Database).
SR19: The system shall allow users to disable animations that may induce photosensitive seizures.
SR20: The system shall ensure that the Tutorial / Hint Manager is synchronized with the current game state.
SR21: The system shall remind the user how long they have been playing every hour.

7 Roadmap

This section outlines which safety and security requirements will be implemented as part of the capstone timeline, and which safety and security requirements will

be deferred. The requirements implemented as part of the capstone timeline are considered high priority requirements.

7.1 Safety and Security Requirements for Capstone Timeline

- SR1: It is important that our user data is secure as we are planning to release the game on the internet within the capstone timeline.
- SR2: Similar to SR1, it is important that the user accounts are secure as we are planning to release the game on the internet within the capstone timeline.
- SR5: An important game mechanic is saving the game state to ensure that the application is convenient for users.
- SR6: Similar to SR5, saving the game state is an important mechanic and ensures that the application allows users to exit and resume play.
- SR7: Ensuring that a save state is valid before loading is crucial to ensure that the game does not crash the game if the save is corrupted.
- SR8: An important goal in our game is user engagement. If a generated task is duplicated, user satisfaction could be significantly decreased.
- SR9: If a task is not solvable, user frustration and confusion will increase. This increases the chance that users will abandon the game.
- SR12: If a solution does not correctly solve the current task, user frustration and confusion will increase. This increases the chance that users will abandon the game.
- SR13: Because we are planning to release the game on the internet within the capstone timeline, it is important that we protect our users from harmful language.
- SR14: Notifying the user if their hardware specifications do not meet minimum requirements is important because it gives the user an actionable fix if they are experiencing performance issues with the game.
- SR15: Incorrect highlighting can increase user frustration and confusion, leading to decreased user engagement, satisfaction, and education.
- SR16: By allowing users to confirm parsed data before submitting, the chance of the user submitting parsed information that they did not intend decreases. In turn, this improves user satisfaction.
- SR17: By informing users when input events are not recognized, users can better understand what is going wrong. This will reduce user confusion.

- SR18: One of the main features for the capstone timeline is competitive statistics and leaderboards. Therefore, it is imperitive that these statistics/leaderboards are accurately displayed.
- SR19: Because the application will be released to the public within the capstone timeline, it is imperative that this version of the application takes into account the risk of photosensitive seizures.
- SR20: Effective education of octal arithmetic is an important goal of the application. To achieve this goal in the capstone timeline, the Tutorial / Hint Manager must show up to date hints / tutorial steps based on the game's state.

7.2 Deferred Safety and Security Requirements

- SR3: Although creating a backup is important, due to the recreational nature of the application, it is not a significant detriment if data is lost or corrupted.
- SR4: Similar to SR3, because the data stored is not entirely important (other than it being secure), it is not our main priority to prevent data loss or corruption.
- SR10: Although general rules regarding how the leaderboards are modified will be set up during the capstone timeline, it is not immediately critical to stop malicious users from modifying competitive statistics / leaderboard results.
- SR11: This requirement is not possible without SR3, which itself is deferred. The reasoning for this requirement being deferred follows the same reasoning as SR3.
- SR21: Although addiction can be a concern, it cannot be prioritized over more prevelent risks such as photosensitive seizures. Additionally, addiction-prevention features like this already exist externally (e.g., screen time controls), so there are other methods in place to control usage if necessary while this requirement is deferred.

8 Appendix — Generative AI Use Disclosure

8.1 AI Tools Used

Microsoft Copilot version 155.0 was used in support of researching and reviewing this document.

8.2 How and Where Used

First, Generative AI was useful in reading the assignment description and our current version of the document to provide recommendations for the FMEA table and Safety and Security Requirements. This use affected the FMEA and Safety and Security Requirements sections.

Finally, the main use for Generative AI was receiving the project description and rubric as well as a draft of the document and suggesting additions and/or modifications. In this context, Generative AI was used as a reviewer for the document with the task of checking high-level concepts such as meeting all the rubric requirements and document structure/clarity. This document was originally written manually using original ideas and thinking. Once completed, the document, along with the guidelines for the document, was entered. This approach was generally used for the entire document, section by section, to collect feedback.

8.3 Verification of AI-Generated Content

The suggestions provided by Microsoft Copilot were not implemented directly. Instead, to ensure the content was reasonable/quality and fit the tone and intent of our document, the comments were reviewed by a team member and then implemented manually. Generative AI was not used to create whole paragraphs or generate facts independently.

9 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

Jacob Brodersen

1. The Hazard Definition section went well while writing this deliverable. In particular, I found that the examples and lecture slides were very helpful in creating our own hazard definition.

   Another section that went well was the Safety and Security requirements section. I found that after writing the FMEA table, the associated requirements became clear.

   Finally, I thought that writing the background section went well. This went well because much of it was already defined in the problem statement document. Additionally, feedback from the Problem Statement document and from our supervisor, Dr. Rapoport, allowed us to refine the background and adapt it for this document.

2. One pain point that I experienced during this deliverable was creating the FMEA table. This was difficult and time-consuming because it involved a lot of brainstorming. At first, I found myself trying to think of how a full row would come together before starting on the idea. This became

cumbersome and stifled my creativity. To resolve this issue, I decided to simply focus on coming up with failure modes by themselves. This allowed me to brainstorm at a high level without getting bogged down on the specific details of the failure mode.

Another pain point I experienced during this deliverable was figuring out how to connect the various sections together to make an interconnected document. When looking at the document as a whole, it was very overwhelming, and I was not sure how I was going to make the different sections consistent. To solve this, I found it useful to focus on one section at a time. For example, I made sure that I had all of my components finalized and well-defined before making the FMEA table.

3. Before this deliverable, our team initially anticipated the health-related risk, inducing photosensitive seizures. We also considered the risk of leaking sensitive information like users' passwords. Another risk we thought of beforehand was the risk of losing educational value as education is one of the main goals of our application. Finally, our application being a game, we foresaw the risk of losing user engagement, satisfaction, and/or trust.

   During this deliverable, we thought of risks related to data loss or corruption. These risks came about by talking with our TA during the informal meeting. Another risk that we found during the Hazard Analysis process was risks relating to the game state, such as losing game state data or loading the incorrect game state. We realized the importance of this risk after looking at the RLCatan sample Hazard Analysis. With strong parallels to our game-based project, we found that the risks associated with the Game State Manager were similar. As such, we adapted aspects from that example project to fit our program/application. Finally, through brainstorming, we came up with an additional health hazard related to mental health with bullying / hateful language.

4. One significant risk other than physical harm is leakage of sensitive data. This is important to consider because this risk becoming a reality can lead to loss of trust if it is sensitive user data. Furthermore, sensitive data leakage can result in more serious consequences (e.g., financial, identity theft, fraud, etc.).

   A second risk other than physical harm that is particularly important to our project is the risk of losing educational value. One of the main goals of our project is to provide an educational aspect to our game for octal arithmetic. A risk is that our game does not achieve this goal or even worse, does the opposite and hurts the users' knowledge related to octal arithmetic. This risk is important to consider because if it were to become a reality, a main selling point of our game would become void or could even become a liability of our application.

   Finally, separate from physical harm is harm to mental well-being. For example, one of the hazards we identified was bullying or hateful comments

through the use of usernames. This is important because as creators, we are responsible for helping safeguard our users from not only physical harm, but mental harm as well.

Josh Engel

1. I worked on the SRS-IEEE document and so I dont really have any comments on what went smoothly specifically when writing this hazard analysis.

2. I worked on the SRS-IEEE document and so I dont really have any comments on what pain points were experienced specifically when writing this hazard analysis.

3. I believe this is also covered in Jacob's reflection, but prior to this deliverable we expected the health related risk of photosensitive seizures. As well as security risks that are caused by handling user account data. Then throughout the deliverable we also found the risks of data loss and corruption as well as state loss within the game. These were found through conversations with our supervisor and TA.

4. One other large risk to consider when using any software is data leakage, no matter how sensitive the data this can cause many issues for those using your softwate and particularly when handling sensitive information it is crucially important to ensure that none of that data is leaked to avoid issues such as identity theft or financial schemes. Another risk to consider with software is computational errors. Though this could lead to physical harm is a computer gives the wrong output it can also lead to false alarms through many fields such as finance or truly lead to the loss or unwanted modification of sensitive data due to issues such as integer overflow.

Nehan Munir

1. I worked on the SRS-IEEE document and so I dont really have any comments on what went smoothly specifically when writing this hazard analysis.

2. I worked on the SRS-IEEE document and so I dont really have any comments on what pain points were experienced specifically when writing this hazard analysis.

3. Before this milestone, we were wondering what risks can a game really have, when the examples brought up in class were all about Capstone projects with real consequences for messing up (such as ones helping the medical industry [failure can lead to misdiagnosis and at worst, death] or a mechatronic project used to map caves [losing the robot due to errant control leads to tangible financial detriments]). During our TA meeting, we humourously brought this up to up to our TA, who suggested

thinking about old videogames. Old videogames have a seizure warning before loading into the main screen, due to flashing lights. As our game was browser-based with a very memory-intensive 3D cube visulization, a memory leak can easily cause the 3D visualization to mess up and cause flickering (seen with older graphic cards).

4. As stated before, memory leakage is a very important issue to cover as the project should never surpass the browser heap limit (essentially 'browser RAM'), as exceeding it (due to not terminating instances that aren't needed before [ex. imagine the project is running every instance of the 3D visualization locally saved]). This can lead to flickering and while seizures are a physical harm, causing browsers to crash can lead to losing unsaved progress on other tabs. Another worry is cybersecurity as DDoS attacks might not be able to extract well protected data (such as user data), it can still flood the browser deployment, resulting in financial costs as the DNS server provider is trying to respond to every DNS query.